Real-time Intercom Schematic

E77-400MBL-01 LoRa Module + INMP441 + MAX98357A

1. System Architecture

TX Mode (PTT pressed): INMP441 Mic -> I2S -> E77 Module -> LoRa TX
RX Mode (PTT released): E77 Module (LoRa RX) -> I2S -> MAX98357A -> Speaker

2. E77 Module Pin Assignment

Function	Pin	Name	Alt Func	Connection
I2S BCLK	30	PA5	SPI1_SCK	INMP441_SCK + MAX98357_BCLK
I2S LRCK	11	PB12	I2S2_WS	INMP441_WS + MAX98357_LRCK
I2S DOUT	14	PA12	TIM1_ETR	INMP441_DATA -> E77
I2S DIN	12	PA10	TIM1_CH3	E77 -> MAX98357_DIN
PTT Key (K2)	28	PA1	TIM2_CH2/GPIO	TX/RX Toggle
UART TX	19	TXD	Low-power UART	External MCU_RX
UART RX	20	RXD	Low-power UART	External MCU_TX
3.3V	1, 18	VCC	Power	All modules VDD
GND	6, 17	GND	Ground	Common ground

3. Power Network (3.3V)

VCC (3.3V): E77 Pin1/Pin18 -> INMP441 VDD -> MAX98357A VDD -> 10uF capacitor
GND: E77 Pin6/Pin17 -> INMP441 GND -> MAX98357A GND -> 100nF filter x3

4. I2S Audio Bus Connection

Signal	E77 Pin	INMP441	MAX98357A
BCLK (Bit Clock)	PA5 (Pin30)	SCK	BCLK
LRCK (Word Select)	PB12 (Pin11)	WS	LRCK
DATA_IN (Mic)	PA12 (Pin14)	DATA (OUT)	-
DATA_OUT (Amp)	PA10 (Pin12)	-	DIN

5. PTT Button (K2 on E77 Board)

Button	E77 Pin	Function	Description
K2 (PTT)	PA1 (Pin28)	TX/RX Toggle	Press and hold to TX, release to RX

Button Circuit: K2 on E77 board connected to PA1 (Pin28), using built-in pull-up

6. Module Configuration

INMP441: L/R pin -> GND (Left channel address 0x48)
MAX98357A: GAIN -> GND (Default 9dB gain), SD -> 3.3V (Enable)

7. BOM (Bill of Materials)

Ref	Specification	Qty	Description
E77	E77-400MBL-01	1	LoRa wireless module (with K2 PTT button)
U1	INMP441 module	1	Digital MEMS mic
U2	MAX98357A module	1	I2S amplifier
C1, C5	10uF/10V tantalum	2	Power storage
C2-C4	100nF SMD capacitor	3	Decoupling
SPK	8 Ohm / 1W speaker	1	Audio output

8. PCB Layout Guidelines

Routing Priority:
1. I2S signals (BCLK, LRCK, DATA) - Length mismatch < 5mil
2. Power traces - 3.3V (20mil), GND (30mil)
3. Button control lines - Standard (10mil)

Notes:
- Place mic near PCB edge
- Keep LoRa antenna away from audio signals
- Amplifier output away from sensitive signals
